

**DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)**

CYP81

PROJECT WORK

PROJECT SYNOPSIS

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PROJECT TEAM MEMBERS

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Title of the Project: Real-Time Deepfake Generation and Detection Immune System

Project Stream: Visual Intelligence & Cybersecurity (AI/ML + Communication & Networks)

Abstract

This project is a real-time deepfake "immune system" that generates and detects synthetic media to stay ahead of new threats. It uses an adversarial training loop. A generative adversarial network (GAN) creates realistic deepfakes while a detection engine learns to spot them in video, audio, and metadata streams. The system provides live authenticity scores, cryptographic evidence, and easy integration with social media, conferencing platforms, and enterprise security tools. By combining visual intelligence with network-level threat monitoring, this project offers scalable, privacy-friendly protection against deepfake-driven misinformation, financial fraud, and identity theft.

Problem Statement: Deepfake technology enables the creation of hyper-realistic synthetic videos and audio that can mislead the public, manipulate elections, defame individuals, and aid cybercrime. Existing detection tools are often static, slow, or easily bypassed. There is a critical need for a self-updating, real-time detection system capable of adapting to new deepfake techniques while maintaining high accuracy and scalability across platforms.

Objective:

- Adversarial Training Ecosystem – Develop a dual-loop architecture where a GAN deepfake generator and a detection model continuously challenge and improve each other.
- Real-Time Multimodal Detection – Implement video, audio, and metadata forensics with frame-level temporal and biometric consistency checks to flag deepfakes instantly.
- Secure Evidence & Integration – Provide cryptographic proof, content authentication, and APIs for integration with social media, video-conferencing, and enterprise SIEM systems.
- Edge Deployment & Privacy – Optimize models for on-device/edge inference using TensorFlow Lite and ONNX to preserve user privacy and ensure low-latency detection.

Scope of the Project:

1. Societal Impact
 - Protects individuals and organizations from misinformation, impersonation, and financial fraud.
 - Builds public trust in digital media and online communication.
 - Supplies verifiable evidence to law enforcement and courts.
2. Sustainability
 - Edge computing reduces cloud-compute costs and energy use.
 - Open, modular design encourages continuous community-driven improvement.
3. Market Analysis
 - Deepfake detection market projected to exceed \$5 B by 2030.
 - Applicable to finance, media, government, healthcare, and education sectors needing real-time identity verification.
4. Innovation

- First end-to-end “immune system” combining adversarial deepfake generation, detection, and automated threat intelligence.
- Multimodal analysis with cryptographically verifiable results for legal-grade forensics.

What contribution to the society would the project make?

This project strengthens the integrity of information ecosystems by preventing the spread of harmful synthetic media. It safeguards personal reputations, democratic processes, and financial systems, while fostering responsible AI development through continuous research and open standards.

Hardware & Software Requirements:

Hardware Requirements

- Development/Training Server: Multi-GPU workstation ($\geq 2 \times$ NVIDIA RTX A6000 or similar, 64 GB GPU memory total), 128 GB RAM, ≥ 2 TB SSD
- Edge Deployment Devices: Standard smartphones or edge servers with ≥ 8 GB RAM and ARM/Intel CPU + optional GPU.

Software Requirements

- Core AI: Python 3.11, PyTorch, TensorFlow Lite, ONNX
- Computer Vision: OpenCV
- Frontend (Demo/Tool): ReactJS (web) or React Native (mobile)
- Backend/API: FastAPI/Flask
- Edge Deployment: Use Android/iOS for mobile, Linux for server/dev
- Security: OpenSSL for cryptographic operations
- Storage: Local SQLite or cloud-integrated



Guide Comments:

Signature of the Guide with date

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